



NanoCodec: Towards High-Quality Ultra Fast Speech LLM Inference

Edresson Casanova¹, Paarth Neekhara¹, Ryan Langman¹, Shehzeen Hussain¹,
Subhankar Ghosh¹, Xuesong Yang¹, Ante Jukić¹, Jason Li¹, Boris Ginsburg¹

¹NVIDIA Corporation, USA

{ecasanova, pneekhara, rlangman, shehzeenh, subhankarg, xueyang, ajukic, jasoli, bginsburg}@nvidia.com

Abstract

Large Language Models (LLMs) have significantly advanced audio processing by leveraging audio codecs to discretize audio into tokens, enabling the application of language modeling techniques to speech data. However, existing audio codecs often operate at high frame rates, leading to slow training and inference, particularly for autoregressive models. To address this, there is growing interest in low frame-rate audio codecs, which reduce the number of autoregressive steps required to generate one second of audio. In this paper, we conduct ablation studies to examine the impact of frame rate, bitrate, and causality on codec reconstruction quality. Based on our findings, we introduce NanoCodec, a state-of-the-art audio codec that achieves high-quality compression at just 12.5 frames per second (FPS). NanoCodec outperforms related works across various bitrate ranges, establishing a new benchmark for low-latency and efficient Speech LLM training and inference.

Index Terms: Neural Audio Codec, Speech Codec, Speech LLMs, Speech Synthesis, Text-to-Speech

1. Introduction

Audio coding is an important signal processing technique, that compresses audio signals into discrete codes and then uses these codes to reconstruct the original audio. These technologies have been central to fields such as audio transmission and communication [1, 2]. Recently, audio codecs have been applied to various downstream tasks. For instance, discrete codes generated by audio codecs combined with large language models (LLMs) have shown impressive results in tasks like zero-shot text-to-speech (ZS-TTS) [3, 4] and speech-to-speech translation (S2ST) [5, 6].

In recent years, Neural Audio Codecs (NACs), which process raw waveforms as both input and output, have emerged as a promising solution that balances audio quality and bitrate [7, 8, 9, 2]. Notable examples include SoundStream [7], Encoder [8], DAC [10], and APCodec [1]. These models utilize Residual Vector Quantization (RVQ) [11] to encode audio at low bitrates while preserving fidelity, often incorporating generative adversarial network (GAN)-based losses to enhance reconstruction quality. Research efforts have primarily focused on improving quantization strategies to enhance NAC performance. Several studies have refined RVQ techniques [12] and explored alternative approaches such as Finite Scalar Quantization (FSQ) [13, 9, 2]. In addition, task-specific optimizations have been introduced, including integration of semantic information into quantization [3]. Further improvements have been achieved through architectural modifications [14] and reductions in computational complexity [15], collectively advancing the efficiency and effectiveness of NACs. Additionally, re-

searchers have pursued significant reductions in bitrate. For instance, WavTokenizer [16] encoded 24kHz audio at 0.9 kbps using a single codebook at 75 FPS, achieving audio quality comparable to previous NACs at a similar bitrate. TAAE [17] encodes 16 kHz audio at 25 FPS with bitrates of 0.4 and 0.7 kbps, surpassing previous works trained at the same bitrate. TS3-Codec [18] encodes 16 kHz audio at 40-50 FPS with 0.68kbps-0.85kbps, achieving significantly better audio quality and computational efficiency than previous models at a similar bitrate.

Recently, there has been growing interest in reducing the frame rate of NACs, aiming to reduce the number of autoregressive steps required to generate audio in Speech LLMs. Notable examples include Mimi [19], a model designed for duplex speech-to-speech applications, which compresses 24 kHz audio at 1.1 kbps and 12.5 FPS while maintaining reasonable audio quality. The Low Frame-rate Speech Codec (LFSC) [2] provides high-quality 22.05 kHz audio compression at 1.89 kbps with a frame rate of 21.5 FPS. More recently, LFSC was integrated into the Koel-TTS model [4], resulting in a threefold acceleration in inference for autoregressive zero-shot speech synthesis, while achieving state-of-the-art (SOTA) zero-shot performance in intelligibility, speaker similarity, and audio quality. This highlights the significant advantages of frame rate reduction.

Despite notable advancements in the field, several recent studies have overlooked the critical need for low latency, which presents significant challenges for enabling real-time streamable inference [1, 14, 2]. Achieving low latency in Speech LLMs can be accomplished through frame rate reduction [2], architectural optimizations [18], and the incorporation of a causal decoder [19], which facilitates inference without requiring token/frame caching. In this paper, we conduct a series of ablation studies to investigate the effects of frame rate, bitrate, and causality on codec reconstruction quality. Based on these insights, we propose NanoCodec, a SOTA audio codec that delivers high-quality compression at a remarkably low frame rate of 12.5 FPS, while outperforming related works across various bitrate ranges.

The key contributions of this work are as follows:

- We introduce *NanoCodec*, a novel state-of-the-art, partially causal audio codec that delivers high-quality audio compression at an exceptionally low frame rate of 12.5 FPS.
- We conduct extensive ablation studies on frame rate reduction, causal audio coding, and the impact of bitrate on our proposed codec.
- To demonstrate our codec’s effectiveness in enhancing audio quality and reducing inference time for Speech LLMs, we train and evaluate a SOTA LLM-based TTS model.
- Our codec implementation is publicly available in the NeMo

repository¹, and the pretrained checkpoints can be accessed on Hugging Face².

The audio samples for each of our experiments are available on the demo website³.

2. NanoCodec

NanoCodec builds upon the LFSC [2], incorporating architectural modifications to enhance audio quality and intelligibility, while reducing the model number of parameters. The model consists of a fully convolutional generator network and three discriminators.

The generator includes an encoder, vector quantization, and a HiFi-GAN-based decoder [20]. The encoder features five residual blocks [20], each comprising three residual layers similar to the multi-receptive field fusion (MRF) module proposed by [20]. Unlike [2], which used a dilation rate of 1, we adopted dilation rates of 1, 3, and 5, as suggested by [9]. We also explored various frame rates for the codec. To achieve a frame rate of 21.5, as in [2], each residual block is followed by a 1D convolutional layer with strides of [2, 2, 4, 8, 8]. For frame rates of 25, 12.5, and 6.25, we used strides of [2, 3, 3, 7, 7], [2, 3, 6, 7, 7], and [3, 4, 6, 7, 7], respectively. The encoder has 24 initial channels, which are doubled after each downsampling 1D convolutional layer, and has a total of 30.4M parameters.

The decoder, based on the HiFi-GAN vocoder, uses upsampling rates that are the reverse of the encoder’s 1D convolutional strides, following the approach in [2]. However, our decoder has 864 initial channels, which are halved after each upsampling layer. Inspired by [21], we replaced the Leaky ReLU activation in the decoder with Snake activation [22]. The decoder contains 31.6M parameters.

Additionally, we explored a causal variant of the encoder and decoder by replacing noncausal convolutional layers with causal ones, as in [19]. In total, these changes reduce the generator’s parameter count by 45% compared to LFSC.

For the discriminators, we employ three neural networks using squared-GAN and feature-matching loss, consistent with [2]. We retained the WavLM-based discriminator and the multi-period discriminator [20] from the original work [2], but replaced the multi-scale complex STFT discriminator [8] with the multi-band multi-scale STFT discriminator [10].

In addition, inspired by [23], we incorporate Speaker Consistency Loss (SCL) to improve speaker similarity in reconstructed speech. Specifically, we employ a pre-trained speaker encoder [24] to extract speaker embeddings from both the generated and ground truth audio, maximizing their cosine similarity. Formally, let $\phi(\cdot)$ denote the function that outputs a speaker embedding, cos_sim the cosine similarity function, α a positive real-valued scaling factor controlling the contribution of SCL to the overall loss, and n the batch size. The SCL is then defined as:

$$L_{\text{SCL}} = \frac{-\alpha}{n} \sum_{i=1}^n \text{cos_sim}(\phi(g_i), \phi(h_i)), \quad (1)$$

where g_i and h_i correspond to the ground truth and generated audio, respectively. In our experiments, we set $\alpha = 0.1$.

For vector quantization, we followed [2], using Finite Scalar Quantization (FSQ) with eight codebooks, each containing four dimensions. The codebook levels are set to [8, 7, 6,

6], resulting in a total of 2016 codes per codebook. Additionally, we investigated various configurations of codebook numbers and levels to meet the specific requirements of our ablation studies.

2.1. Datasets

For codec training, we utilize the two 22.05kHz audio datasets used in [2]. The Common Voice derived [25] training set comprises 105 languages, totaling 2.7 million utterances, and 3.2k hours of audio from about one-hundred thousand speakers. The MLS English training dataset consists of 6.2 million utterances and 25.5k hours of audio from 4329 speakers.

2.2. Training setup

We trained our codec for approximately 196,000 steps using 48 A100 GPUs with a batch size of 32. The total accumulated batch size was 1,536, and the model processed roughly 301 million samples. The training was conducted on 1.1-second audio excerpts. We used the Adam optimizer [26] for both the generator and the discriminator, with $\beta_1 = 0.8$, $\beta_2 = 0.99$, and an initial learning rate of $2e-4$, which decayed exponentially with a gamma of 0.998.

2.3. Results and Discussion

We followed an evaluation strategy similar to [2]. To assess perceptual quality, we estimate Mean Opinion Scores (MOS) using Torchaudio-Squim [27] (SQMOS) and the wide-band Perceptual Evaluation of Speech Quality (PESQ) metric [28]. Spectral accuracy is evaluated by computing the L1 distance between log mel-spectrograms (Mel Dist.). To quantify the intelligibility of the codec’s reconstruction, we compute the Character Error Rate (CER) by comparing transcriptions from the Massively Multilingual Speech (MMS) model [29] for both the ground-truth and reconstructed audio. Speaker similarity is assessed using the Speaker Encoder Cosine Similarity (SECS) [23], computed with the SOTA ECAPA2 speaker encoder [30].

For evaluation, we use the 44.1kHz MLS test set [2], which includes 200 samples per language across eight languages. Additionally, we assess performance on the F10 and M10 speakers from the DAPS clean dataset to evaluate model quality on studio-recorded audio.

2.3.1. Ablation Study

We conduct an ablation study to analyze the enhancements introduced to the LFSC architecture, as well as the effects of causality, frame rate, and bitrate reduction on our codec. Table 1 presents the evaluation results on the 44.1kHz MLS test set and the F10 and M10 DAPS speakers. For consistency, both ground-truth and reconstructed audio were downsampled to 16kHz during metric computation.

Architectural Improvements: The proposed architectural modifications enhanced the LFSC model across nearly all metrics while reducing the number of parameters by approximately 45%.

Causality: A causal codec is preferable for Speech LLM models as it minimizes first-token latency. In contrast, the non-causal LFSC requires a lookahead of approximately five frames (232.5 ms), meaning that tokens from the first five autoregressive steps must be cached to achieve the same quality in both streaming and offline inference. This constraint introduces a significant delay in generating the initial audio output. For a demonstration of the impact of this lookahead requirement,

¹<https://github.com/NVIDIA/NeMo>

²<https://huggingface.co/nvidia/nanocodec-22khz-1.78kbps-12.5fps>

³<https://edresson.github.io/NanoCodec>

Table 1: *Reconstruction performance of different codec configurations using MLS 44.1kHz test set and F10 and M10 DAPS speakers*

Codec	Sampling rate	Bitrate	Token/Sec	Frames/Sec	# Codebooks	# Codes	Encoder	Decoder	Dataset	SQMOS (↑)	PESQ(↑)	Mel Dist. (↓)	SECS (↑)	CER(↓)
LFSC[2]	22.05kHz	1.89kbps	172	21.5	8	2016	Noncausal	Noncausal	MLS DAPS	4.432 4.677	2.831 3.122	0.146 0.141	0.841 0.807	2.529 0.655
Ours	22.05kHz	1.89kbps	172	21.5	8	2016	Noncausal	Noncausal	MLS DAPS	4.427 4.693	2.919 3.271	0.136 0.130	0.876 0.844	2.203 0.577
Ours	22.05kHz	1.1kbps	100	12.5	8	2016	Causal	Causal	MLS DAPS	4.394 4.655	1.926 2.058	0.211 0.2157	0.735 0.717	5.490 1.124
Ours	22.05kHz	1.1kbps	100	12.5	8	2016	Noncausal	Noncausal	MLS DAPS	4.423 4.683	2.592 2.893	0.158 0.154	0.825 0.785	3.614 0.927
Ours	22.05kHz	1.1kbps	100	12.5	8	2016	Noncausal	Causal	MLS DAPS	4.426 4.564	2.378 2.646	0.177 0.171	0.788 0.777	3.617 0.839
Ours	22.05kHz	1.1kbps	100	25	4	2016	Noncausal	Causal	MLS DAPS	4.415 4.662	2.534 2.863	0.167 0.160	0.798 0.755	3.749 0.622
Ours	22.05kHz	1.1kbps	100	12.5	8	2016	Noncausal	Causal	MLS DAPS	4.426 4.564	2.378 2.646	0.177 0.171	0.788 0.777	3.617 0.839
Ours	22.05kHz	1.1kbps	100	6.25	16	2016	Noncausal	Causal	MLS DAPS	4.400 4.652	2.081 2.135	0.200 0.213	0.746 0.711	5.293 1.419
Ours	22.05kHz	1.78kbps	162.5	12.5	13	2016	Noncausal	Causal	MLS DAPS	4.441 4.697	2.760 3.030	0.143 0.139	0.862 0.831	2.423 0.758
Ours	22.05kHz	1.1kbps	100	12.5	8	2016	Noncausal	Causal	MLS DAPS	4.426 4.564	2.378 2.646	0.177 0.171	0.788 0.777	3.617 0.839
Ours	22.05kHz	0.8kbps	50	12.5	4	65536	Noncausal	Causal	MLS DAPS	4.369 4.617	2.033 2.237	0.205 0.202	0.706 0.645	6.519 1.189
Ours	22.05kHz	0.6kbps	50	12.5	4	4032	Noncausal	Causal	MLS DAPS	4.407 4.662	2.012 2.205	0.205 0.204	0.701 0.656	7.792 1.469
Ours	22.05kHz	1.78kbps	162.5	12.5	13	2016	Noncausal	Causal	MLS DAPS	4.441 4.697	2.760 3.030	0.143 0.139	0.862 0.831	2.423 0.758
Ours	22.05kHz	1.89kbps	172	21.5	8	2016	Noncausal	Causal	MLS DAPS	4.427 4.666	2.837 3.156	0.150 0.145	0.854 0.832	2.434 0.601

please refer to our demo website⁴. To address this issue, we explored different combinations of causal and non-causal encoder/decoder configurations. Following the Mimi codec [19], we reduced the bitrate to 1.1 kbps and the frame rate to 12.5 FPS. Our fully causal model yielded the worst results, particularly in intelligibility and speaker similarity. Conversely, the fully non-causal model achieved the best performance across nearly all metrics. The partially causal model, which employs a non-causal encoder and a causal decoder, demonstrated competitive performance compared to the fully non-causal model. For certain Speech LLM tasks, such as TTS and Speech-to-Speech models where ASR or SSL models serve as the encoder, a causal encoder is unnecessary since only the codec decoder is used during inference. In these scenarios, our experiments suggest that a partially causal codec may be beneficial. Given that our primary goal in this paper is TTS downstream task, we will adopt the partially causal model for future experiments.

Frame Rate Reduction: Reducing the frame rate in Speech LLMs decreases the number of autoregressive steps per second of audio, improving inference speed and reducing latency. To evaluate this effect, we trained our 1.1 kbps codec at 25, 12.5, and 6.25 FPS. While the 25 FPS codec outperformed the 12.5 FPS variant in PESQ and Mel Distance, both exhibited comparable performance in perceptual quality (SQ-MOS), speaker similarity (SECS), and intelligibility (CER). In contrast, the 6.25 FPS codec showed a significant decline, particularly in intelligibility. We hypothesize that at 6.25 FPS, the model frequently compresses two distinct phonemes within a single frame, whereas such occurrences are considerably less frequent at 12.5 FPS. Given that most English speakers produce speech at a rate of approximately ten to twelve phonemes per second [31], this observation is further supported.

Bitrate Reduction: Recent research has extensively explored bitrate reduction in neural audio codecs. The primary hypothesis is that lower bitrates yield simpler representations, making it easier for Speech LLMs to learn the target distribution. However, there is an inherent trade-off between bitrate, audio quality, and intelligibility. To further investigate this trade-off, we trained our 12.5 FPS codec at 1.78 kbps, 1.1 kbps, 0.8 kbps, and 0.6 kbps. As expected, higher-bitrate models produced better results across all metrics. Notably, bitrate reduc-

tion has a substantial impact on intelligibility (CER), whereas general audio quality and speaker similarity are comparatively less affected. Specifically, reducing the bitrate from 1.78 kbps to 1.1 kbps led to a relative degradation of approximately 1.49 times in CER (from 2.423 to 3.617). A further reduction to 0.8 kbps resulted in a 2.69 times relative increase in CER (from 2.423 to 6.519). At 0.6 kbps, intelligibility deteriorated even further, with pronunciation errors becoming significantly more frequent. These findings strongly suggest that such a decline in intelligibility could severely impact the performance of Speech LLM models trained with extremely low-bitrate codecs.

21.5 FPS vs. 12.5 FPS: We further compared our model trained at 1.78 kbps with a frame rate of 12.5 FPS against the same model trained at 1.89 kbps with a frame rate of 21.5 FPS. Both models show comparable performance, with 12.5 FPS excelling in SQMOS and 21.5 FPS in PESQ. Speaker similarity and intelligibility remain comparable, indicating that lower frame rates can maintain quality while improving efficiency.

2.3.2. Comparison with Related Works

To ensure a fair comparison with related works, we trained our model at a bitrate comparable to previous approaches. For LFSC [2], Mimi [19], WavTokenizer [16], and TAAE [17], we utilized publicly available checkpoints and inference code. For TS3-Codec [18], we evaluated samples generously provided by the authors. For consistency, we downsampled both ground-truth and reconstructed audio to 16kHz during metric computation, ensuring a fair comparison with TAAE and TS3-Codec, which operate at this sampling rate. Table 2 presents the results of our evaluation on the 44.1 kHz MLS test set and the F10 and M10 DAPS speakers, highlighting the performance of our approach relative to state-of-the-art codecs.

Our model, trained at 1.78 kbps, outperforms the 1.89 kbps LFSC across most metrics, while benefiting from the advantages of a causal decoder and approximately 45% fewer parameters. At 1.1 kbps, our model outperforms the Mimi codec across all metrics, showing a particularly large improvement in intelligibility. At 0.8 kbps, while the overall audio quality of our model is slightly lower than that of the 0.9 kbps WavTokenizer, it consistently outperforms the WavTokenizer across all other evaluation metrics. When compared to the 0.85 kbps TS3-Codec, our 0.8 kbps model exhibits better intelligibility on both

⁴<https://edresson.github.io/NanoCodec>

Table 2: Reconstruction performance of different SOTA codecs.

Codec	Bitrate	Dataset	SQMOS	PESQ	Mel Dist.	SECS	CER
LFSC [2]	1.89kbps	MLS	4.432	2.831	0.146	0.841	2.529
		DAPS	4.677	3.122	0.141	0.807	0.655
Ours	1.78kbps	MLS	4.441	2.760	0.143	0.862	2.423
		DAPS	4.697	3.030	0.139	0.831	0.758
Mimi [19]	1.1kbps	MLS	4.333	2.132	0.238	0.678	7.221
		DAPS	4.302	2.316	0.244	0.569	1.629
Ours	1.1kbps	MLS	4.426	2.378	0.177	0.788	3.617
		DAPS	4.564	2.646	0.171	0.777	0.839
WavTokenizer [16]	0.9kbps	MLS	4.404	1.877	0.207	0.576	15.27
		DAPS	4.671	2.222	0.213	0.535	4.042
TS3-Codec [18]	0.85kbps	MLS	4.392	2.389	0.183	0.672	8.159
		DAPS	4.674	2.770	0.167	0.733	1.522
Ours	0.8kbps	MLS	4.369	2.033	0.205	0.706	6.519
		DAPS	4.617	2.237	0.202	0.645	1.189
TAAE [17]	0.7kbps	MLS	4.370	2.053	0.320	0.323	18.355
		DAPS	4.611	2.250	0.248	0.372	3.692
Ours	0.6kbps	MLS	4.407	2.012	0.205	0.701	7.792
		DAPS	4.662	2.205	0.204	0.656	1.469

evaluation datasets. Additionally, it achieves superior speaker similarity on the multilingual MLS dataset, though it shows a slight reduction in speaker similarity for two high-quality English speakers in the DAPS dataset. Despite these minor trade-offs in specific metrics, our model delivers these results with a 50 bps smaller bitrate and is trained at 22.05 kHz, while the TS3-Codec was trained at 16 kHz. This distinction is significant, as our model encodes a broader frequency range. At 0.6 kbps, our model significantly outperforms the TAAE codec in most metrics, although TAAE slightly surpasses our model in PESQ. Overall, our model achieves SOTA results across a wide range of bitrates, demonstrating the efficiency and effectiveness of the proposed NanoCodec. Notably, it excels in enhancing key metrics such as intelligibility and speaker similarity, further solidifying its superiority in low-bitrate audio coding scenarios.

3. Zero-Shot TTS study

3.1. Experiments setup

To assess the performance of our codec in comparison to the LFSC and explore the effects of bitrate reduction, we adopted Koel-TTS [4], a SOTA LLM-based TTS model. Among Koel-TTS architectures, we use the Decoder Context variant with Classifier-Free Guidance (CFG) as it achieves the best performance in zero-shot TTS. For simplicity, we did not apply Direct Preference Optimization in our experiments. For training the TTS model, we used the same datasets as [4]. These datasets consist of 18,000 hours of English data from three sources: the train-clean 100 and 360 subsets of LibriTTS [32], HiFiTTS [33], and a 17,000-hour subset of the LibriVox MLS dataset [34].

The model was trained using 1.89 kbps LFSC, as well as NanoCodec at 1.89 kbps, 1.78 kbps, and 1.1 kbps bitrates, using a fixed five-second context from an alternate utterance of the same speaker. Training ran for 200,000 steps on 32 A100 GPUs with a batch size of 8, totaling 256. The AdamW optimizer was used with a 1e-4 learning rate. During inference, we applied multinomial Top-k sampling with $K = 80$, CFG scale 2.5, and a temperature of 0.6.

3.2. Results and Discussion

To evaluate the performance of the Koel-TTS model with various codecs, we followed the methodology proposed by [4]. We evaluated the model on unseen speakers, using the same subset of *test-clean* LibriTTS containing 180 utterances from a total of 36 speakers. We use a random 5 second slice from the context audio during inference for all experiments. CER was computed using the Parakeet-TDT ASR model [35]. SECS was calculated using the Titanet-Small [36] speaker verification model.

For MOS evaluation, each sample was rated on a 5-point scale by at least 37 independent listeners. Additionally, we measured the inference real-time factor (RTF) and time-to-first-audio (TTFA) using a batch size of 16 on a single RTX 6000 GPU. To ensure consistency, both RTF and TTFA values were normalized by setting the fastest model’s values to 1 and scaling those of the other models accordingly.

Table 3: Koel-TTS evaluation using different codecs

Codec	Bitrate	Frames/Sec	MOS(↑)	CER(↓)	SECS(↑)	RTF(↓)	TTFA(↓)
LFSC [4]	1.89kbps	21.5	4.17 ± 0.04	0.85±0.35	0.719	2.48	2.85
Ours	1.89kbps	21.5	4.16 ± 0.04	0.76±0.28	0.738	2.47	1.14
Ours	1.78kbps	12.5	4.01 ± 0.04	0.49±0.03	0.635	1.06	1.01
Ours	1.1kbps	12.5	4.02 ± 0.04	0.69±0.12	0.565	1	1
Ours - 10s context	1.78kbps	12.5	3.84 ± 0.04	0.55 ± 0.07	0.691	1.06	1.01

* Note that RTF and TTFA values are normalized by setting the smallest value to 1.

Table 3 presents the results of our ZS-TTS experiments. Koel-TTS trained with the 21.5 FPS NanoCodec achieved CER and MOS values comparable to those of the model trained with 21.5 FPS LFSC, while exhibiting better speaker similarity, a similar real-time factor, and a latency (TTFA) approximately 2.5 times shorter. In contrast, the model trained with the 12.5 FPS 1.78 kbps NanoCodec showed better intelligibility than its 21.5 FPS counterpart, while improving the real-time factor by 2.33 times and slightly reducing TTFA. However, it struggled to generalize to unseen speakers, leading to lower speaker similarity, and reduced overall quality, despite improving intelligibility. A similar trend was observed in the Koel-TTS model trained with the 12.5 FPS 1.1 kbps NanoCodec, which exhibited even lower speaker similarity and a slightly higher CER. This degradation is likely due to the reduced reconstruction quality of the 1.1 kbps NanoCodec compared to its 1.78 kbps variant.

We hypothesize that the performance gap in speaker similarity and overall quality between the 12.5 FPS and 21.5 FPS models arises because the 12.5 FPS model generates nearly twice as much speech per autoregressive step, making the task more challenging. Conversely, the lower CER observed with the 12.5 FPS model could be attributed to the phoneme input and codec tokens having a similar rate, making alignment learning easier. Additionally, the shorter sequence length reduces the available speaker context by a factor of 1.7, potentially impairing speaker similarity. To mitigate this issue, we extended the context audio length from 5 to 10 seconds through repetition, which consistently improved speaker similarity from 0.635 to 0.691, but at the cost of a slight degradation in overall quality. These findings suggest that fully harnessing the potential of the 12.5 FPS NanoCodec may require architectural refinements to Koel-TTS.

4. Conclusions and future work

In this work, we introduced NanoCodec, a state-of-the-art audio codec that achieves high-quality compression at a bitrate of 1.78 kbps and a frame rate of 12.5 frames per second. Experimental results demonstrate that NanoCodec surpasses existing approaches at the same bitrate, offering significantly higher intelligibility and speaker similarity. Furthermore, we showed that a Speech LLM trained with NanoCodec achieves quality comparable to that of a model trained with the LFSC, while exhibiting a lower real-time factor and reduced speech output latency, benefiting from our causal decoder. Our ablation studies further reveal that bitrate reduction does not always improve downstream performance due to the trade-off between quality and intelligibility. In future work, we plan to further investigate architectural modifications to Koel-TTS to fully leverage the potential of the 12.5 FPS NanoCodec.

5. References

- [1] Y. Ai, X.-H. Jiang, Y.-X. Lu, H.-P. Du, and Z.-H. Ling, "Apcodec: A neural audio codec with parallel amplitude and phase spectrum encoding and decoding," *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, vol. 32, p. 3256–3269, jun 2024. [Online]. Available: <https://doi.org/10.1109/TASLP.2024.3417347>
- [2] E. Casanova, R. Langman, P. Neekhara, S. Hussain, J. Li, S. Ghosh, A. Jukić, and S.-g. Lee, "Low frame-rate speech codec: a codec designed for fast high-quality speech llm training and inference," *ICASSP*, 2025.
- [3] X. Zhang, D. Zhang, S. Li, Y. Zhou, and X. Qiu, "Speechtokenizer: Unified speech tokenizer for speech large language models," *arXiv preprint arXiv:2308.16692*, 2023.
- [4] S. Hussain, P. Neekhara, X. Yang, E. Casanova, S. Ghosh, M. T. Desta, R. Fejgin, R. Valle, and J. Li, "Koel-tts: Enhancing llm based speech generation with preference alignment and classifier free guidance," 2025. [Online]. Available: <https://arxiv.org/abs/2502.05236>
- [5] X. Li, Y. Jia, and C.-C. Chiu, "Textless direct speech-to-speech translation with discrete speech representation," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [6] M. Kim, J. Choi, D. Kim, and Y. M. Ro, "Many-to-many spoken language translation via unified speech and text representation learning with unit-to-unit translation," *arXiv preprint arXiv:2308.01831*, 2023.
- [7] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "Soundstream: An end-to-end neural audio codec," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.
- [8] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *arXiv preprint arXiv:2210.13438*, 2022.
- [9] R. Langman, A. Jukić, K. Dhawan, N. R. Koluguri, and B. Ginsburg, "Spectral codecs: Spectrogram-based audio codecs for high quality speech synthesis," *arXiv preprint arXiv:2406.05298*, 2024.
- [10] R. Kumar, P. Seetharaman, A. Luebs, I. Kumar, and K. Kumar, "High-fidelity audio compression with improved rvqgan," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [11] A. Vasuki and P. Vanathi, "A review of vector quantization techniques," *IEEE Potentials*, vol. 25, no. 4, pp. 39–47, 2006.
- [12] Y. Luo, J. Yu, H. Chen, R. Gu, and C. Weng, "Gull: A generative multifunctional audio codec," *arXiv preprint arXiv:2404.04947*, 2024.
- [13] F. Mentzer, D. Minnen, E. Agustsson, and M. Tschannen, "Finite scalar quantization: Vq-vae made simple," *arXiv preprint arXiv:2309.15505*, 2023.
- [14] L. Xu, J. Jiang, D. Zhang, X. Xia, L. Chen, Y. Xiao, P. Ding, S. Song, S. Yin, and F. Sohel, "An intra-brnn and gb-rvq based end-to-end neural audio codec," *arXiv preprint arXiv:2402.01271*, 2024.
- [15] L. Xu, J. Wang, J. Zhang, and X. Xie, "Lightcodec: A high fidelity neural audio codec with low computation complexity," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 586–590.
- [16] S. Ji, Z. Jiang, X. Cheng, Y. Chen, M. Fang, J. Zuo, Q. Yang, R. Li, Z. Zhang, X. Yang *et al.*, "Wavtokenizer: an efficient acoustic discrete codec tokenizer for audio language modeling," *arXiv preprint arXiv:2408.16532*, 2024.
- [17] J. D. Parker, A. Smirnov, J. Pons, C. Carr, Z. Zukowski, Z. Evans, and X. Liu, "Scaling transformers for low-bitrate high-quality speech coding," *arXiv preprint arXiv:2411.19842*, 2024.
- [18] H. Wu, N. Kanda, S. E. Eskimez, and J. Li, "Ts3-codec: Transformer-based simple streaming single codec," *arXiv preprint arXiv:2411.18803*, 2024.
- [19] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour, "Moshi: a speech-text foundation model for real-time dialogue," *arXiv preprint arXiv:2410.00037*, 2024.
- [20] J. Kong, J. Kim, and J. Bae, "Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis," *Advances in neural information processing systems*, vol. 33, pp. 17 022–17 033, 2020.
- [21] S.-g. Lee, W. Ping, B. Ginsburg, B. Catanzaro, and S. Yoon, "Bigvgan: A universal neural vocoder with large-scale training," *arXiv preprint arXiv:2206.04658*, 2022.
- [22] L. Ziyin, T. Hartwig, and M. Ueda, "Neural networks fail to learn periodic functions and how to fix it," *Advances in Neural Information Processing Systems*, vol. 33, pp. 1583–1594, 2020.
- [23] E. Casanova, J. Weber, C. D. Shulby, A. C. Junior, E. Gölge, and M. A. Ponti, "Yourtts: Towards zero-shot multi-speaker tts and zero-shot voice conversion for everyone," in *International Conference on Machine Learning*. PMLR, 2022, pp. 2709–2720.
- [24] H. S. Heo, B.-J. Lee, J. Huh, and J. S. Chung, "Clova baseline system for the voxceleb speaker recognition challenge 2020," *arXiv preprint arXiv:2009.14153*, 2020.
- [25] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, and G. Weber, "Common voice: A massively-multilingual speech corpus," in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 2020, pp. 4218–4222.
- [26] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [27] A. Kumar, K. Tan, Z. Ni, P. Manocha, X. Zhang, E. Henderson, and B. Xu, "Torchaudio-squim: Reference-less speech quality and intelligibility measures in torchaudio," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [28] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, "Perceptual evaluation of speech quality (pesq)-a new method for speech quality assessment of telephone networks and codecs," in *2001 IEEE international conference on acoustics, speech, and signal processing. Proceedings (Cat. No. 01CH37221)*, vol. 2. IEEE, 2001, pp. 749–752.
- [29] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi *et al.*, "Scaling speech technology to 1,000+ languages," *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.
- [30] J. Thienpondt and K. Demuynck, "Ecapa2: A hybrid neural network architecture and training strategy for robust speaker embeddings," *arXiv preprint arXiv:2401.08342*, 2024.
- [31] P. Roach, *A little encyclopaedia of phonetics*. Verlag nicht ermittelbar, 2002.
- [32] H. Zen, V. Dang, R. Clark, Y. Zhang, R. J. Weiss, Y. Jia, Z. Chen, and Y. Wu, "Libritts: A corpus derived from librispeech for text-to-speech," *Interspeech 2019*, 2019.
- [33] E. Bakhturina, V. Lavrukhin, B. Ginsburg, and Y. Zhang, "Hi-Fi Multi-Speaker English TTS Dataset," in *Proc. Interspeech 2021*, 2021, pp. 2776–2780.
- [34] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, and R. Collobert, "Mls: A large-scale multilingual dataset for speech research," *Interspeech 2020*, 2020.
- [35] H. Xu, F. Jia, S. Majumdar, H. Huang, S. Watanabe, and B. Ginsburg, "Efficient sequence transduction by jointly predicting tokens and durations," in *International Conference on Machine Learning*. PMLR, 2023.
- [36] N. R. Koluguri, T. Park, and B. Ginsburg, "Titanet: Neural model for speaker representation with 1d depth-wise separable convolutions and global context," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 8102–8106.